GALGOTIAS UNIVERSITY

SCHOOL OF ENGINEERING & TECHNOLOGY (SET)
Department of Computer Science & Engineering

SOFTWARE ENGINEERING LAB REPORT

PROJECT: LOANZO

Institutional Peer-to-Peer (P2P) Lending & Credit Ledger Ecosystem

Lab Modules Covered:
- **Lab 2 Outcome:** Problem Definition Document
- **Lab 3:** Feasibility Study (Technical, Economic, Operational & Constraints)
- **Lab 4:** Requirement Elicitation – I (Stakeholder Identification & Analysis)

Academic Term:	Fall Semester 2026	**Course Code:**	BCSE308P / SE Lab
Subject:	Software Engineering	**Target System:**	Android (Offline-First Clean Arch)
Author / Candidate:	Student Engineering Team	**Submission Date:**	September 2026
Evaluation Status:	Final Lab Record / Defense Ready	**Curriculum Source:**	Galgotias Univ LMS Plan

1. Problem Definition Document

1.1 Project Title & Domain

Title: Loanzo — Institutional P2P Lending, Repayment Ledger & Collateral Management Platform.
Domain: Financial Technology (Fintech), Decentralized Peer-to-Peer Microfinance, Legal Contracts, and Banking-Grade Mobile Security.

1.2 Context & Industry Background

In developing economies like India, over 70% of micro, small, and medium enterprises (MSMEs) and unbanked retail borrowers rely heavily on informal, non-institutional credit. Friends, relatives, community lenders, and local merchants facilitate loans worth billions of rupees annually. However, this informal credit system operates almost exclusively on unrecorded paper chits, unorganized messaging threads, and verbal trust.

1.3 Concrete Problem Statement

The existing informal credit and unorganized P2P lending ecosystem suffers from severe structural vulnerabilities:

- **Lack of Legally Enforceable Agreements:** Informal personal and business loans are executed without structured promissory contracts. When disputes arise, lenders possess zero legally admissible evidence under the Indian Evidence Act or Information Technology Act 2000 Section 10A.
- **Default Risk & Disputed Repayment Tracking:** Borrowers and lenders frequently disagree on outstanding principal, accumulated interest, partial tranche disbursements, and repayment dates due to messy paper notebooks.
- **Fraudulent Identity & Collateral Manipulation:** Without authentic government KYC verification (Aadhaar/PAN) or physical asset inspection, lenders risk financing fraudulent borrowers or ghost collateral.
- **Predatory Usury & Unregulated Recovery:** Unregulated local moneylenders charge exorbitant, compounding interest rates (often 36%–120% per annum) with aggressive, undocumented recovery tactics.
- **Device Vulnerability & Mobile Session Tampering:** Conventional fintech apps either force frustrating, continuous re-logins or leave sensitive financial ledgers exposed when users switch apps to copy an SMS OTP.

1.4 Proposed Solution: The Loanzo Platform

Loanzo solves these systemic challenges by engineering an offline-first, mobile-native financial ledger application that formalizes informal credit without the bureaucracy of traditional commercial banks. Core solutions include:

- **Unified P2P Bilateral Dashboard:** Single interface where any user can act as both Lender and Borrower simultaneously.
- **On-Device Legal Contract Generation & Dual e-Signing:** Creates tamper-evident, multi-page PDF loan agreements with Aadhaar e-Sign, ML Kit liveness selfie capture, and cryptographic hashes.
- **Government Verified KYC & Document Vault:** Integrated DigiLocker, PAN OCR, and AES-256 encrypted storage for identity credentials.
- **Empaneled Doorstep Field Agent Network:** Institutional physical inspection workflow for on-site collateral appraisal and borrower verification.
- **Banking-Grade Session Security:** 3-minute background multi-tasking grace window, non-destructive biometric quick-unlock, and 2-day hard session expiry.

1.5 Objectives & Scope

Primary Objectives: (1) Zero-ambiguity bilateral loan ledger; (2) Legally binding digital contracts in seconds; (3) 100% offline readability via Room SQLite; (4) Banking-level data privacy adhering to RBI P2P guidelines.

Scope Boundaries: The system facilitates loan matching, legal agreements, collateral verification, and repayment tracking. Loanzo does not hold retail banking deposits or operate escrow accounts directly; actual monetary settlements occur via UPI/IMPS with cryptographic receipt verification.

2. Lab 3: Feasibility Study Report

A rigorous feasibility study was conducted across Technical, Economic, and Operational dimensions to evaluate project viability before full architectural engineering.

2.1 Technical Feasibility Analysis

The technical feasibility evaluates whether the required technologies, libraries, hardware capabilities, and engineering skills are available and capable of delivering the system invariants.

Component	Selected Technology	Technical Justification & Feasibility Evaluation
UI Toolkit	Jetpack Compose (Material 3)	Declarative reactive UI, elimination of XML fragmentation, 60fps animations, native dark/light theme switching.
Core Language	Kotlin 1.9+ Coroutines & Flow	Type-safe, null-safe, structured concurrency enabling asynchronous background warm-up during splash.
Local Database	Android Room v15 (SQLite Embedded)	Guarantees 100% offline-first functionality. Sub-10ms in-memory cached queries eliminate white screens.
Cloud & Auth	Firebase Auth + Firestore + WorkManager	Seamless bi-directional syncing via SyncWorker queue; resilient against intermittent rural connectivity.
Document Vault	OpenPDF / Android PdfDocument + AES-256	Local, on-device agreement generation with embedded logo and biometric signature without paid cloud APIs.
Biometric Engine	AndroidX Biometric 1.2.0 API	Hardware-backed Fingerprint, Face Unlock, and Device Credential PIN fallback across Android 8.0 to 15+.
Device Security	Hardware Fingerprinting (DeviceSecurityHelper)	SHA-256 cryptographic binding of Android ID and hardware signature to block cloned APK sessions.

Technical Feasibility Conclusion: Highly Feasible. All target tech stacks are production-proven, open-source, and supported natively by modern Android platforms.

2.2 Economic Feasibility Analysis

Economic feasibility assesses the development, deployment, and operational costs against the tangible and intangible financial returns.

Cost Category	Traditional Fintech Core	Loanzo Architecture	Economic Advantage
Server Infrastructure	■45,000 - ■1,20,000 / mo (Dedicated AWS/GCP)	■0 / mo (Free Spark Tier + Serverless Edge)	99% reduction via on-device Room SQLite & WorkManager sync.
Document Generation	■5 - ■15 per agreement (Paid DocuSign / Adobe API)	■0 (Native OpenPDF / Android Canvas engine)	Zero marginal cost per generated loan contract.
Identity Verification	■20 - ■50 per KYC call (Third-party verification)	Direct DigiLocker OAuth & On-Device ML Kit	85% savings utilizing Indian Government public digital rails.
Maintenance & Ops	High (24/7 dedicated DevOps engineers)	Low (Client-heavy Android architecture)	Self-contained APK resilience minimizes server maintenance.

Tangible & Intangible ROI:
• **Monetization Model:** Nominal platform fee (0.5%–1.0%) on completed loan origination, premium certified doorstep inspection fees (split with field agents), and corporate PDF dossier vault exports.
• **Payback Period:** Estimated within 4.5 months of deployment at a modest volume of 2,000 active P2P loans.

2.3 Operational Feasibility Analysis

Operational feasibility determines whether the solution seamlessly integrates into real-world consumer habits, field operations, and legal requirements.

- **End-User Usability (PIECES Framework):** Designed for users with basic smartphone literacy. Incorporates bilingual Devnagari typography (Hindi, Marathi), large touch targets, color-coded status badges, and an executive drill-down hub.
- **Legal & Regulatory Adherence:** Fully compliant with RBI guidelines for peer-to-peer lending platforms (NBFC-P2P Directions 2017) and Information Technology Act 2000 Section 10A guaranteeing legal contract validity for tamper-evident digital signatures.
- **Doorstep Field Agent Operability:** Certified agents use an isolated workflow with GPS location tagging, camera inspection sheets, and real-time duty status toggling without exposing consumer lending/borrowing features.

2.4 Constraints and Assumptions

Identified Technical & Operational Constraints:

1. *Operating System Constraint:* Requires Android 8.0 (API Level 26) or higher for hardware-backed BiometricPrompt and WorkManager APIs.
2. *Device Hardware Constraint:* Physical camera and biometric scanner (fingerprint/face) required for full digital contract execution.
3. *Regulatory Cap:* RBI P2P guidelines mandate aggregate lending caps per individual across all P2P platforms (■50,00,000 limit).

Key Project Assumptions:

1. Users possess valid Indian government identity credentials (Aadhaar number and PAN card).
2. Borrowers and lenders maintain at least periodic internet connectivity for WorkManager queue synchronization.

3. Lab 4: Requirement Elicitation – I

Requirement Elicitation represents the foundational stage of discovering, analyzing, and documenting the explicit needs and subtle pain points of all parties involved in the Loanzo ecosystem.

3.1 Stakeholder Identification & Categorization

Stakeholders were categorized into Primary, Secondary, and Tertiary classes using standard Software Engineering Stakeholder Salience Analysis:

Stakeholder Group	Category	Core Responsibilities & Capabilities	Critical Pain Points & Expectations
Borrower (Consumer / MSME)	Primary	Requests credit, submits KYC & collateral documents, e-signs contracts, pays EMI tranches.	Wants transparent fair interest, zero hidden fees, immediate approval feedback, and dignity in recovery.
Lender (Retail / Angel)	Primary	Offers capital, evaluates borrower risk profile, disburses funds, monitors ledger.	Demands ironclad legal enforceability, verifiable KYC, default alerts, and structured repayment schedules.
Field Inspection Agent	Secondary	Performs on-site doorstep visits, inspects physical collateral, uploads geo-tagged evidence.	Needs simple offline inspection forms, duty toggling, fraud warning tools, and automated payout tracking.
Guarantor / Co-Signer	Secondary	Vouches for borrower credibility, provides secondary repayment pledge, e-signs guarantee pact.	Requires clear disclosure of liability limits and automated notification upon borrower delinquency.

Platform Master Admin	Tertiary	Monitors system health, resolves disputes, approves agent empanelment, audits compliance.	Requires comprehensive audit logs, real-time Telegram bot alerts, and master oversight controls.
Regulatory Authorities (RBI / IT Auditors)	Tertiary	Audits compliance with P2P lending caps, data localization, and IT Act digital contract standards.	Demands complete audit trails, SHA-256 agreement integrity, and zero data leakage to foreign servers.

3.2 Requirement Elicitation Methodologies Used

To ensure complete requirements coverage, a multi-method elicitation framework was executed:

1. **Semi-Structured Interviews:** Conducted with 35 informal retail borrowers, 20 retail lenders, and 8 microfinance recovery executives across urban and semi-urban markets.
2. **Domain Observation & Artifact Analysis:** Analyzed 50+ informal physical promissory notes, pawnshop receipts, and ledger registers (Bahi-Khata) to understand existing informal recording mental models.
3. **Interactive Rapid Prototyping:** Deployed Jetpack Compose interactive wireframes to validate mobile gesture usability, biometric quick-unlock timing, and multi-tranche repayment interfaces.
4. **Regulatory Deconstruction:** Synthesized RBI Master Directions for NBFC-P2P and Section 10A of IT Act 2000 into formal engineering rules.

3.3 Functional Requirements (FR) Specification

The key functional capabilities elicited from stakeholder sessions include:

- **FR-01 (Dual-Role Identity & Authentication):** Unified login allowing users to operate as borrower or lender without multi-account switching. Includes biometric login and forgot-password recovery.
- **FR-02 (Encrypted Document Vault & DigiLocker):** Secure AES-256 storage for Aadhaar, PAN, and salary slips, gated behind password or fingerprint verification before decryption.
- **FR-03 (Bilateral Loan Origination):** Enables borrowers to 'Request Loan' or lenders to 'Grant Loan' with interest rate, tenure, purpose, and counterparty binding by phone or ID.
- **FR-04 (Multi-Tranche Disbursement & Milestone Ledger):** Supports phased disbursement based on project milestones with tranche-specific repayment schedules.
- **FR-05 (On-Device Legal Contract Generation & Dual eSign):** Produces a multi-page legal agreement with app logo, party details, and canvas signature capture.
- **FR-06 (Doorstep Agent Empanelment & Inspection):** Role-isolated portal for certified agents to inspect collateral, verify physical address, and record geo-coordinates.

3.4 Non-Functional Requirements (NFR) Specification

- **NFR-01 (Banking Session Security):** App must auto-lock to biometric shield after 3 minutes of background inactivity while maintaining active screen state. Inactivity > 2 days must trigger hard logout.
- **NFR-02 (Zero-Glitch Startup Warm-up):** Cold startup must leverage the 3.5s cinematic logo animation to pre-warm Room database queries in parallel, ensuring zero visual layout shifts or spinners upon dashboard appearance.
- **NFR-03 (Hardware Cryptographic Binding):** The active session must be anchored to the device hardware signature (`DeviceSecurityHelper`) to prevent cloned APK session hijacking.
- **NFR-04 (Offline-First Resilience):** All loan ledgers, documents, and historical payments must remain viewable with 0% network connectivity.
- **NFR-05 (Multilingual Localization):** Complete native UI localization across English, Hindi, and Marathi without runtime translation lag.

3.5 MoSCoW Prioritization Matrix

Priority Class	Requirement Description	System Impact & Rationalization
MUST HAVE (Critical Core)	Bilateral Loan Origination, DigiLocker KYC, Room Offline DB, On-Device Legal PDF Agreement, Banking Session Auto-Lock (3m/2d).	Mandatory for platform viability, legal contract validity, and financial data security.
SHOULD HAVE (Important)	Doorstep Agent Inspection Workflow, 3.5s Parallel Startup Warmup, Multilingual Devnagari Support, Push Notifications.	Significantly improves user trust, eliminate UI layout jumps, and broadens geographic accessibility.
COULD HAVE (Desirable)	Telegram Bot Integration (@Loanzo_bot), Loan & EMI Simulator Tool, Social Community Marketplace Feed.	Enhances user engagement and administrative alert convenience without impeding core lending.
WON'T HAVE (v1) (Deferred)	Direct Escrow Banking Gateway, Automated Credit Bureau Scraping (CIBIL API), Cross-border Remittance.	Deferred to v2.0 due to NBFC license dependency and third-party commercial API costs.

Academic Submission Endorsement:
This comprehensive documentation for **Lab 2 (Problem Definition)**, **Lab 3 (Feasibility Study)**, and **Lab 4 (Requirement Elicitation – I)** represents an authentic engineering deliverable for the Loanzo P2P Lending System. All technical requirements, security parameters, and feasibility metrics correspond to the production Android implementation.

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